

University of Windsor

MECH 8025 - Automotive NVH Laboratory

Exercise #2

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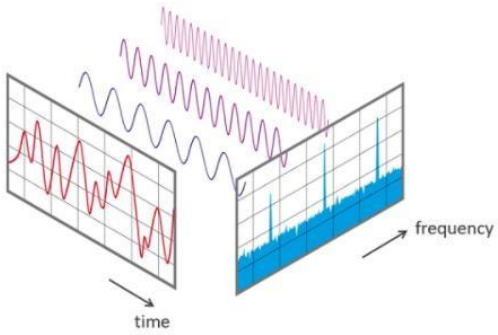
Objectives:

1. **Measure the frequency response of physical systems using one octave and FFT analysis**
2. **Relate the measured frequency response to the physical characteristics of the source**
3. **Become familiar with a noise level measurement procedure**

Abstract

In this Lab experiment, we will study the differences between white noise source and pink source and its pressure level trend with frequency for small speaker and Big speaker. We will try to analyse the effect of big and small speakers on the sound level pressure and judge the best frequencies for each speaker depending on the data given. In addition to this, we will perform FFT analysis considering one octave band in each calculation and will define ideal spectrum of White and pink noise. Moreover, we will draw the hand drill spectrum against Octave band and its sound influence on the overall sound level. Also we will gear mesh frequency to calculate the rpm of the different components.

1. Introduction



View of a signal in the time and frequency domain

In this experiment, we will discuss about the white noise and pink noise and their difference depending on energy parameter. We will also discuss the difference of reading captured in small and big speaker based on size and will derive at conclusion using captured FFT and Octave band data. We will also calculate the RPM of the drill using gear mesh frequency and gear mesh bandwidth.

2. Testing Conditions and Parameters

2.1 Testing 1 Location source & Instruments

- Test Location - Anechoic room
- Noise Source - White and Pink Noise
- Microphone Location - 1 meter from speaker face along centre axis

We will plot the frequency spectrum graph of a white noise and pink noise in an anechoic room. In addition, differences between ideal white noise and pink noise were discussed. Test condition and equipment defined as below.

Instrument details under-mentioned:

Instrument	Manufacturer	Model	Serial Number
FFT Analyser	Bruel & Kjaer	C-Frame	2575990
Software	Bruel & Kjaer	Pulse	N/A
Microphone	Bruel & Kjaer	Type 4189	100303
Preamplifier	Bruel & Kjaer	Type 6969	1058
Misc. Cables	N/A	N/A	N/A
Audio Amplifier	Peavey	MP400	S0530227
Noise Source	Bruel & Kjaer	Pulse	N/A
Large Speaker	Peavey	PR10	E0140441
Small Speaker	Chrysler	Infinity	6593 36670

2.2 Testing 2 Location source & Instruments

- Test Location - CARE Shop
- Noise Source - DeWalt D21009
- Hand Drill Microphone Location - 1 metre from the drill housing

We will measure the frequency response of hand drill by FFT in single Octave band. We will try to judge the physical characteristics of the noise source with the help of extracted data.

Instrument details under-mentioned:

Instrument	Manufacturer	Model	Serial Number
FFT Analyser	Bruel & Kjaer	C-Frame	N/A
Software	Bruel & Kjaer	Pulse	N/A
Microphone	Bruel & Kjaer	Type 4189	100303
Preamplifier	Bruel & Kjaer	Type 6969	1058
Misc. Cables	N/A	N/A	N/A
Hand Drill	DeWalt	D21009	973778

3. Discussion and Solution

In this context, we will discuss about the white and pink noise and their characteristics and reason for such nomenclature of sound. White noise and Pink noise are two types of noises

which are consist of all the audible frequencies that the human ear can hear, but are distinguish depending on the energy flow.

White noise can be explained using the analogy of White color which contains all the wavelength of the light spectrum. Using same analogy of light, we can say that White noise is perfect blend of all audible frequencies whose energy level is equal at all frequencies. In short, White sound includes all frequencies that is audible to the human ear played at an equal intensity or amplitude, and is measured in decibels. Since white noise is comprised of several frequencies, it can often use to mask the other unwanted frequencies. White noise is usually utilized in music creation, either directly or as an input to a filter to generate various types of noise signals.

In contrast to white noise, the pink noise has equal energy in each of the octave band, which means the line of energy level for each octave will have the same level of energy. Due to this, the amplitude at lower octave band is much higher than Octave band compared at higher frequency. It can be also said that the pink noise amplitude decrease with the increase in frequency.

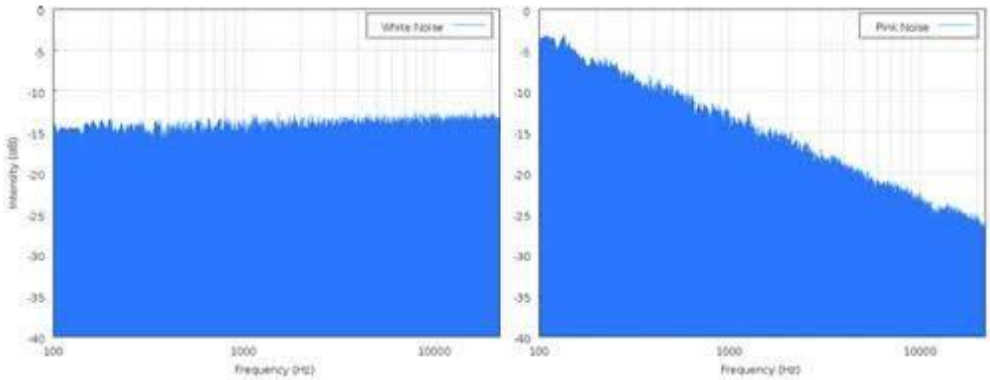


Fig1. FFT analysis of the pink and white noise

Mathematically, white noise is defined equal power per frequency, while pink noise by equal power per octave. Since the lower frequencies are louder than the higher frequencies in pink noise, it sounds less abrasive and thus leads to a better night's sleep.

We found two type of analysis on this. one of them is FFT analysis which is analysed on constant bandwidth whereas Octave analysis on constant bandwidth ratio; thus, the FFT analysis of white noise shows its curve is frequency-independent whereas in the pink noise FFT analysis, it shows a descending curve with frequency.

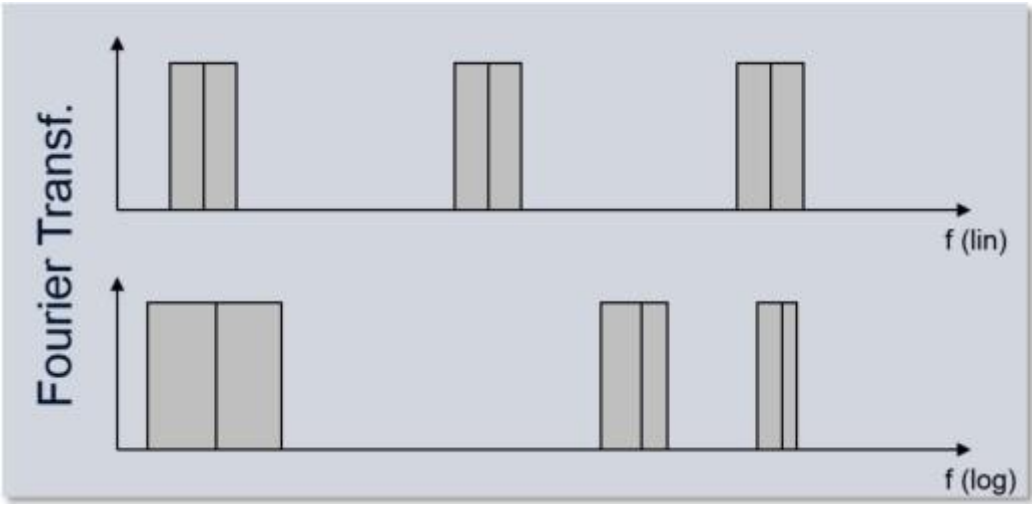


Fig2. Frequency nodes of the FFT analysing on the log and line scale

The frequency distribution in the FFT is shown above. In case of the linear graph frequency, the distribution of frequency nodes is equidistant from each other whereas in the logarithmic frequency, the nodes get closer to each other with increase in the frequency.

Normally we have seen that the FFT results are lower than One Octave results. The reason for this can be related to the energy distribution in FFT and One Octave band. In FFT analysis, the bandwidth is defined hence at higher frequency, resolution works better Wherein in octave filter, the percentage of bandwidth is constant and energy level depend on summation of all the energy level in the octave bandwidth.
As a result, in octave filter the bandwidth rms value increases by increasing frequency. Thus, it shows larger bandwidth as compared to FFT.

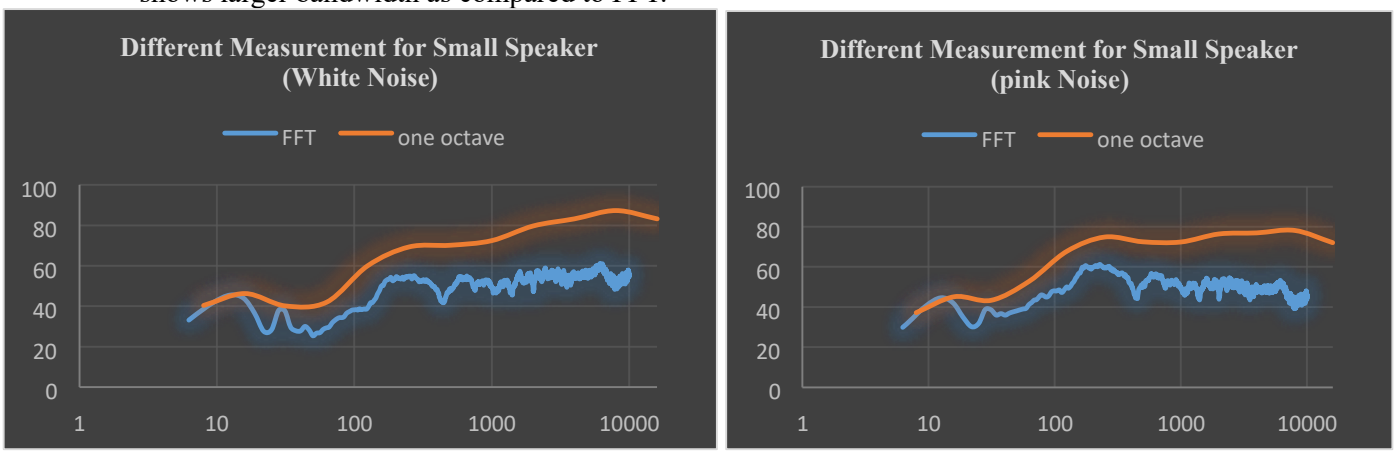


Fig3. Sound Measurement in FFT and One Octave

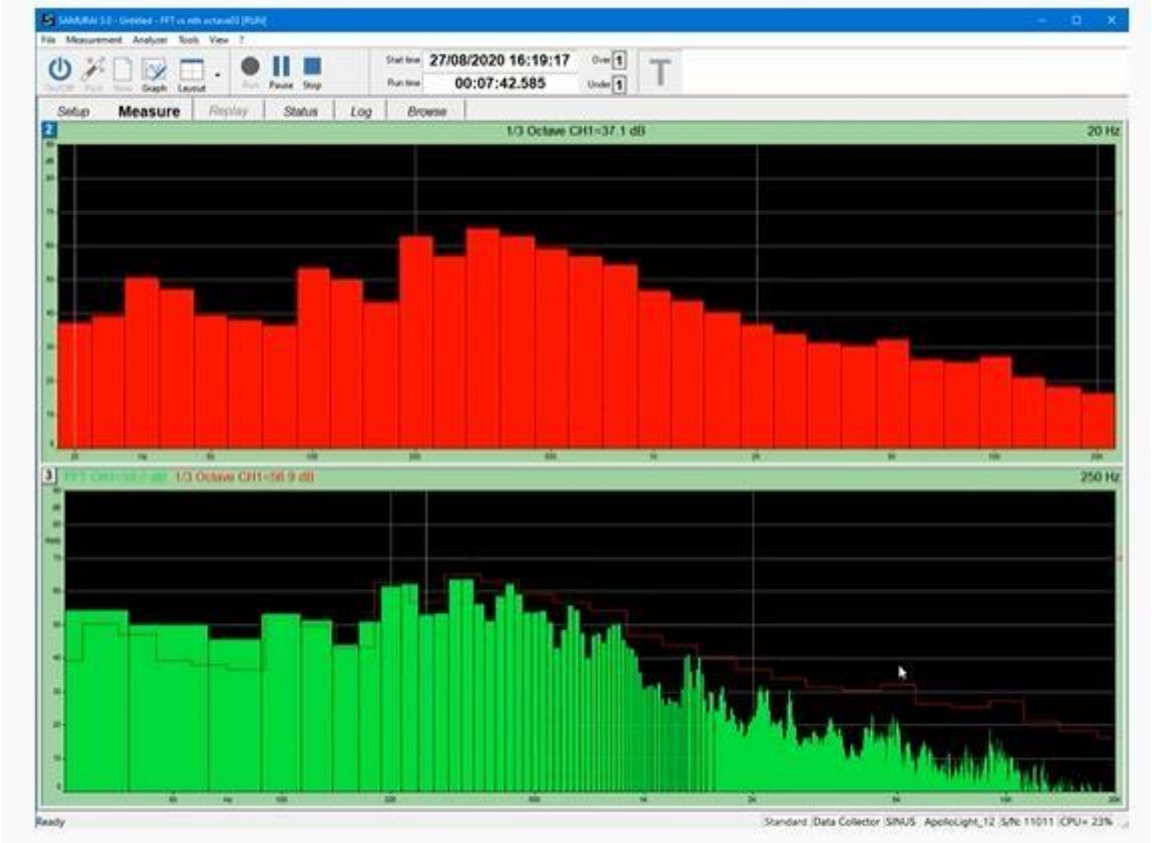


Fig4. Sound Measurement in FFT and One-Third Octave Band Comparison

The pink noise signal is most often used instead of the white noise for observing the speaker frequency response since Pink noise has equal power spectral density which means they have equal energy for each band. At higher frequency, the pink noise becomes lesser power difference level as compared to white noise., thus less abrasive. Human ears can easily respond to such energy bands thus makes it more feasible for human ears to recognize and understand. Thus the pink noise is often used instead of White noise to test audio systems.

Noise is basically Vibration of sound waves that are heard by human ear. It depends on the flow of energy in the air. Big speaker will have bigger diaphragm as compared to smaller ones which means for generation same power level, diaphragm movement in big speaker will be less thus generating low frequency sound or higher wavelength waves as compared to the small speakers. Hence we can say for lower frequencies, big speakers will have better performance as compared to smaller ones and vice-versa for small speakers.

The best suited frequency range is the one which has the lowest variance in sound energy level against frequency. Same is explained in the picture below to relate it to the graph of two speakers against linear frequency output

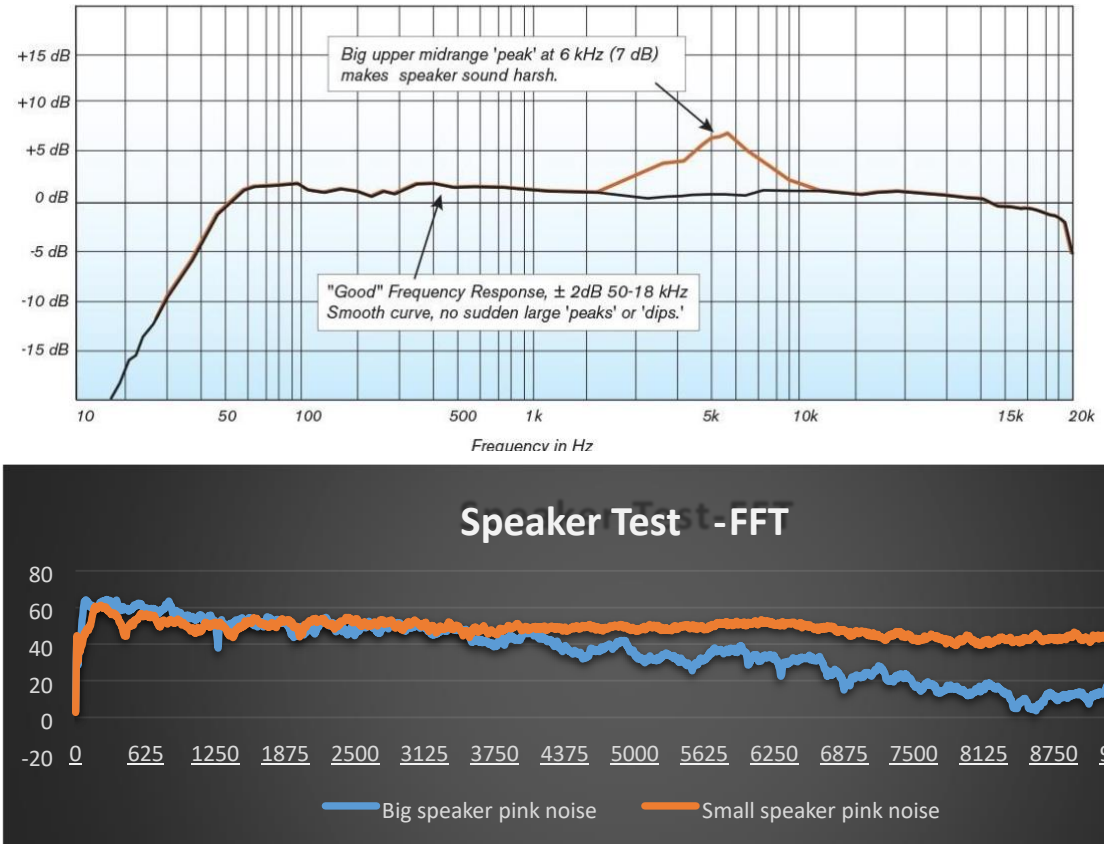


Fig 5. Big and small speaker FFT analysis in linear scale

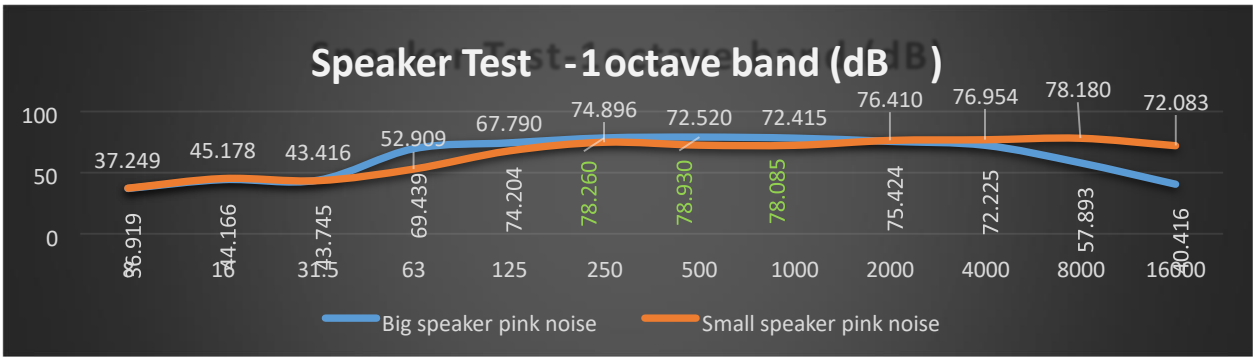


Fig 6. Big and small speaker 1-octave analysis in linear scale

With the help of graphs, we have found that for small speaker, the frequencies which are higher than 2000hz or in between 2000- 6000Hz is best suited as spectrum is more linear and is flat hence small speaker has better sound level in high frequencies, whereas for big speaker, frequencies less than 2000Hz are showing the flat curve, especially, big speaker works extremely good for the frequency between 250-1000 Hz as it has minimal change in sound level and has higher sound level in comparison to small speaker. There is no significant difference between small and large speakers for frequencies less than 31.5 whereas both speakers display variability in the range between 31.5 and 125 Hz.

- **The meaningfulness (usefulness) of the one octave data compared to the FFT data.**

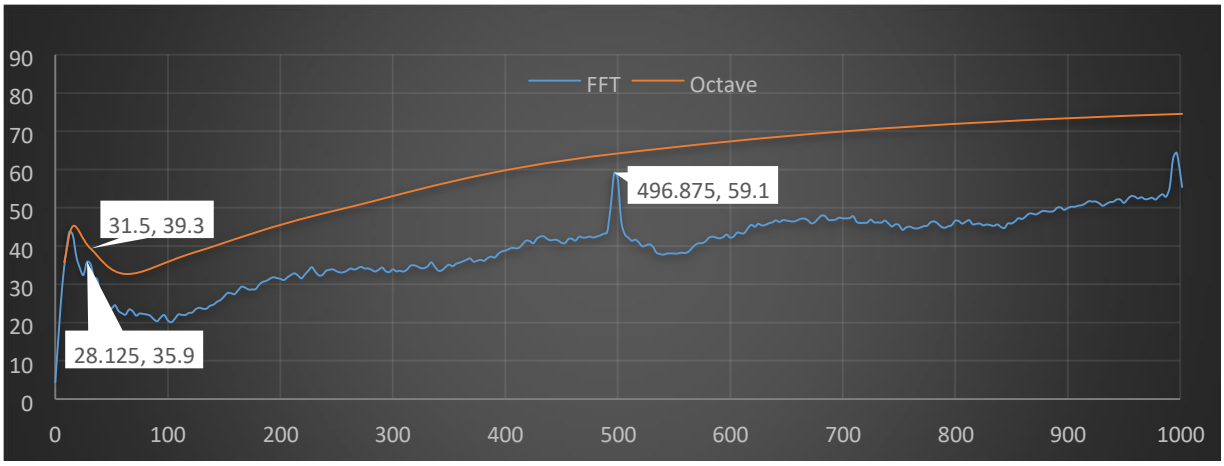


Fig 7. Less difference in very low frequencies (FFT, 1 Octave)

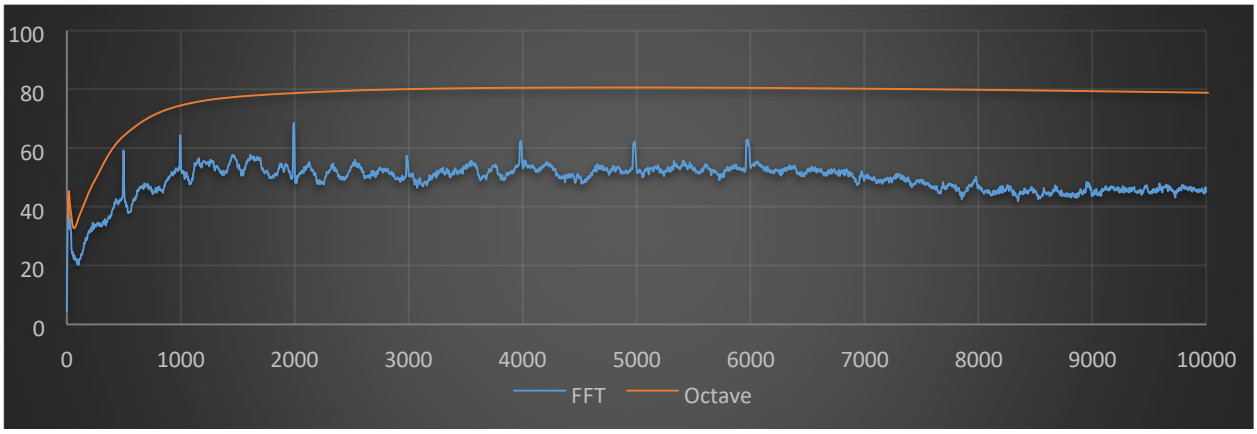


Fig 8. The overall measurements (FFT, 1 Octave)

When frequency increases from 30 Hz, the gap between sound level in two analysis increases. In 106 Hz, sound level for FFT and octave analysis is about 20.1 and 35 dB respectively. Octave band sound level at this frequency is 75 percent higher than FFT. In other frequencies, for FFT, a harmonic chart is observed until about 8000Hz; then, sound level does not change so much. In one octave measurement, after initial rise in sound level, it drops until 63 Hz, then it markedly increases to 2500 Hz and about 80 dB, and then sound level stabilizes at almost 80 dB until the end of the measurement.

Sound level in octave band is higher than FFT due to bandwidth. Bandwidth for FFT analysis is constant 3dB and in octave the ratio of bandwidth stays constant. Consequently, bandwidth increases in octave by increasing frequency hence Octave analysis are better seen in a log scale.

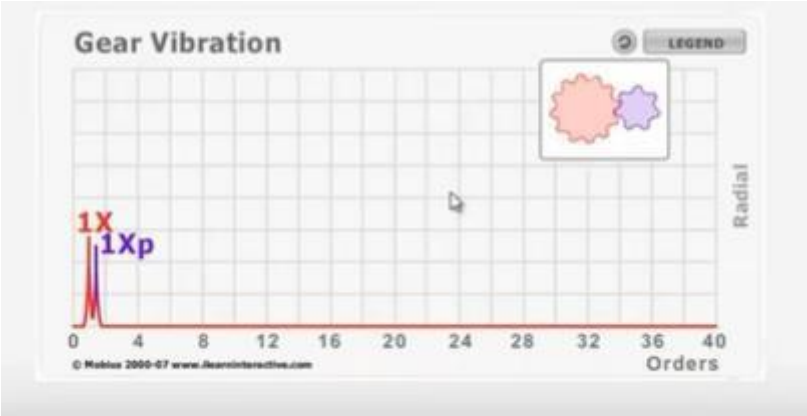
In Figure 8, there is a huge peak in FFT data which is repeated approximately after every 1000 Hz. The mentioned pattern repeats up to 6000 Hz. however, after this point the large peak disappears, and the graph shows a downward fluctuation with small peaks at some frequencies towards higher frequency.

The concept of **Gear Mesh frequency** will help us to calculate RPM and understand the physical characteristics of the source.

Gear Mesh frequency: The rate at which gear and pinion mess with each at regular interval helps to define the tooth-mesh frequency. This mesh frequency is the prime factor in defining the vibration from gearbox and plays important part in noise creation of the object. This frequency is defined by:

$$GMF=ZP\times RPMP=ZG\times RPMG$$

Whenever the two gear mesh with each other at the same speed, they produce their respective vibration from each shaft along with other vibration. There is slight variation in the amplitude of the frequency due to the regular mesh but it is minimal under standard condition. When messing is in progress, there is specific GMF is generate which keeps on repeating under standard conditions. Depending on the gear mesh frequency and amplitude variation, we can try to find the issue in the gear box or messing. The frequency modulation and amplitude modulation will be sign which trigger the issue will the system.





Gear-mesh frequency side bands: Those frequencies which happen after a certain time or are equidistant from the gear-mesh frequency. These sidebands are in sync to the rotating frequency of the pinion and the gear.i.e This sidebands pattern in gear mesh system help to diagnose the performance of gear-pinion assembly and faults if any.

In this case, we will try to solve using the same methodology to solve the problem.

The hand drill consists of four rotating parts located on two main shafts. The primary shaft, including the commutator, motor armature and fan, is connected to the secondary shaft by a

four-tooth helical gear system. From the gear ratio formula, we can derive the velocity of the output shaft i.e (45/4 or 11.25).

The rotating speed of all other components attached to each shaft is equivalent to the speed of the shaft. That is, the commutator, motor armature, and fan rotate at the same speed. However, due to the different number of blades, the frequency of the rod or tooth sound will be different

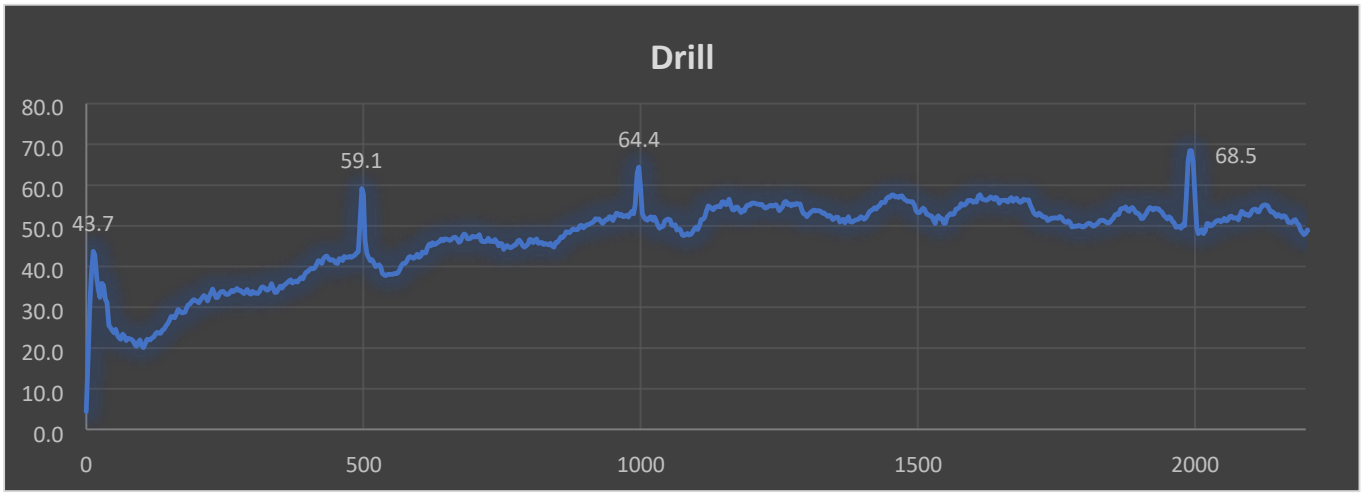


Fig 9. Peak frequencies of Drill (Octave)

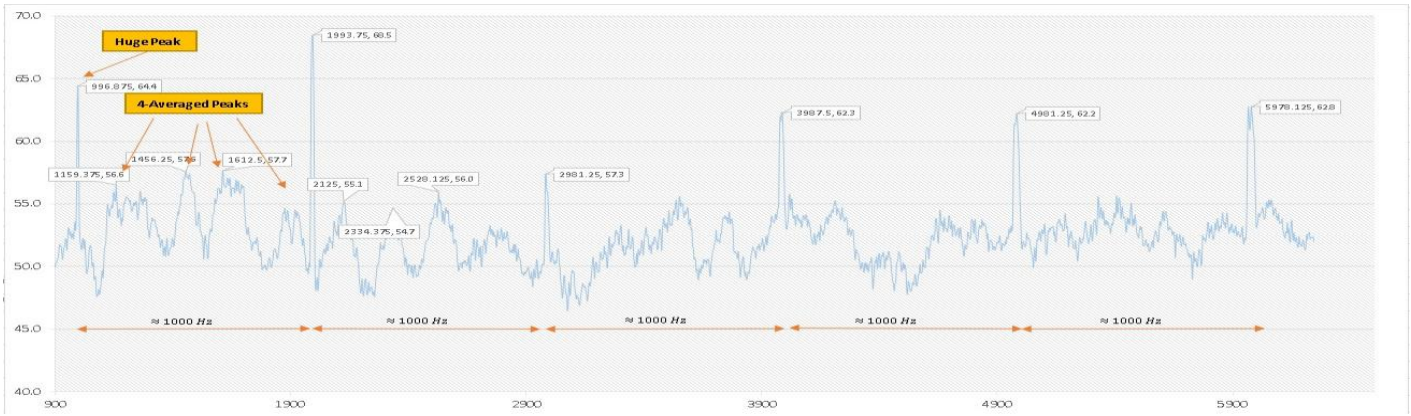


Fig 10. Average Perk Peak frequencies of Drill (Octave)

Normally during the meshing of the Gear system, Driver gear creates first peak in very low frequencies which is 12.5 Hz frequency and then another peak is produced by pinion gear. Second and third peaks in this spectrum (496.875 Hz and 996.875 Hz) belong to the pinion and it follows a certain time period pattern.

Now, Knowing the Pinion frequency and no. of teeth, we can calculate the RPM of the input shaft

$$F_{input\ shaft} = \frac{F_{Pinion}}{Z_p} = \frac{496.875}{4} = \mathbf{124.2\ Hz}$$

Using the input frequency of the shaft, now we can calculate all the frequency of other components:

- $F_{Fan} = F_{input\ shaft} \times Z_f = 124.2 \times 40 = \mathbf{4968.75\ Hz}$
- $F_{commutator} = F_{input\ shaft} \times Z_c = 124.2 \times 24 = \mathbf{2980.8\ Hz}$
- $F_{bush} = F_{input\ shaft} \times Z_b = 124.2 \times 2 = \mathbf{248.4\ Hz}$

The RPM of Hand Drill is equal to the RPM of the input shaft.

Therefore: $\boldsymbol{RPM_{Drill} = 124.2 \times 60 = 7452\ rpm}$

4. Problems

On comparing with the ideal standard graph of pink noise and white noise, there is a lot of deviation in recorded data which means the test recording were not accurate when compared. In additional at 0 Hz, there is noise captured by small speaker and big speakers which means ambient noise was present in the room. The potential problems for this can be due to the orientation of speaker, ambient noise present and error in the FFT calculations.

We already discussed the pattern of White noise and Pink Noise. Looking at the graph of Big speaker and small speaker we can say that do not perform perfectly. Hence the reading error is showing in the graph plotted for White and pink noise thus recording sources have error and have not recorded reading effectively.

Moreover, Ambient sound noises are not considered in this experiment.

5. Conclusion

In this experiment, we have analysed white Noise and pink Noise with the help of Octave and FFT pattern. In white noise, we found it is independent of frequency and is time based, whereas in pink noise we found, the sound level decreases with the increase in frequency. The sound level was recorded in above mentioned noises and investigated with the help of Big and Small speakers.

Octave and FFT methods were two methods considered to analyse the speakers. Overall, the readings are one octave higher than the FFT. At high frequencies or low wavelength, small speakers work better, and at lower frequencies or larger wavelength, larger speakers perform better. The spectrum of white and pink noise is very different from the ideal spectrum because the speaker does not function optimally and its spectrum graph deviates from the standard noise patterns.

The spectrum of the hand drill was drawn by two above mentioned methods were observed. Octave shows a much higher level than the FFT. However, we analysed the FFT and found that there are several peaks in the FFT spectrum. With the use of Gear mesh frequency formula and peak notes in graph, we measured the frequency of the components and the rotational speeds of the various components.